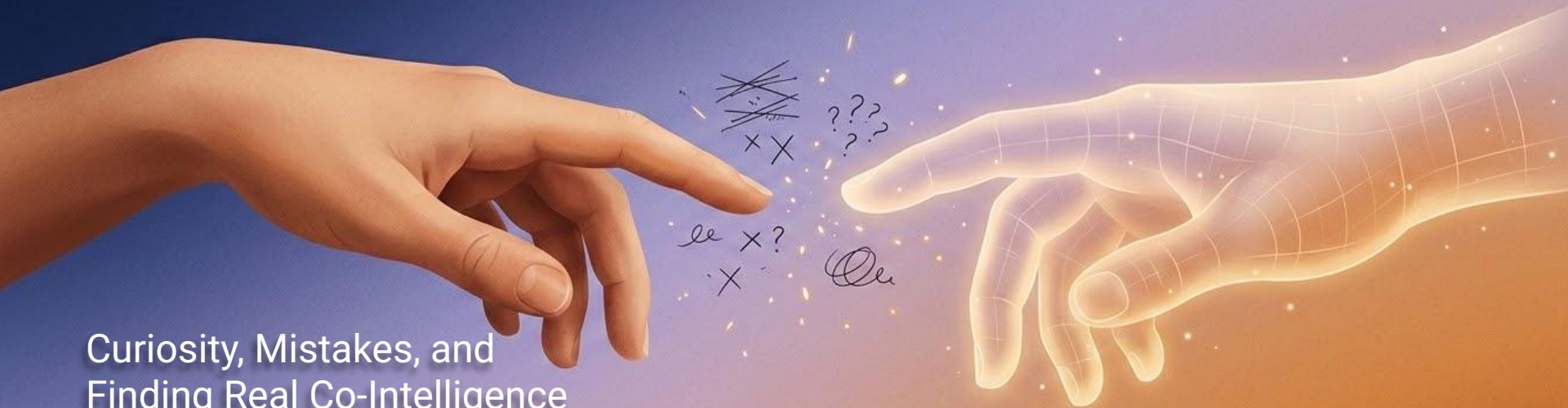


Playing with AI



Curiosity, Mistakes, and
Finding Real Co-Intelligence

Tarek N. Aziz

Associate Professor
NC State University

Curiosity

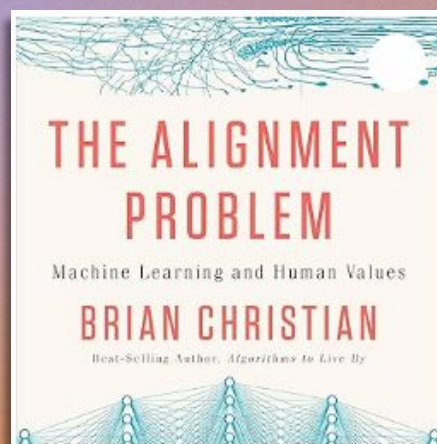
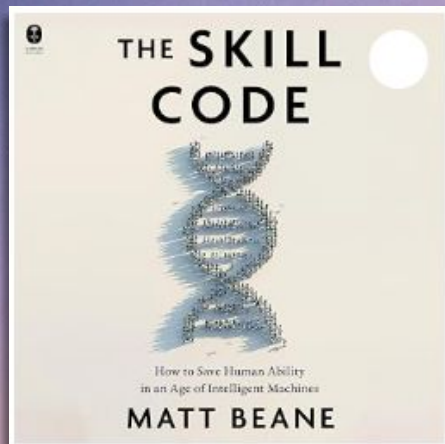
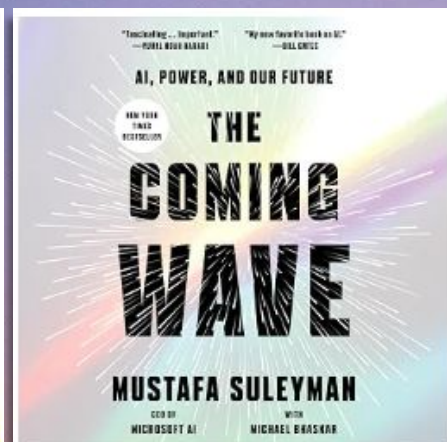
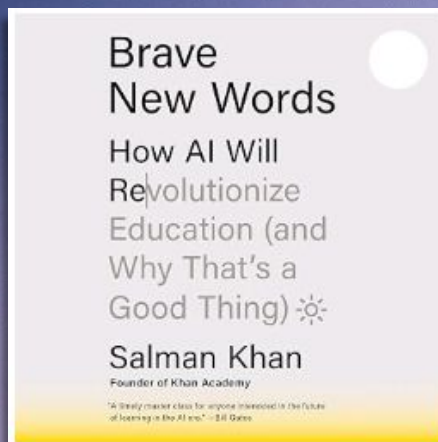
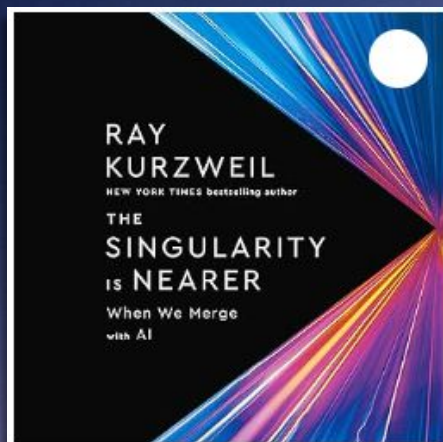
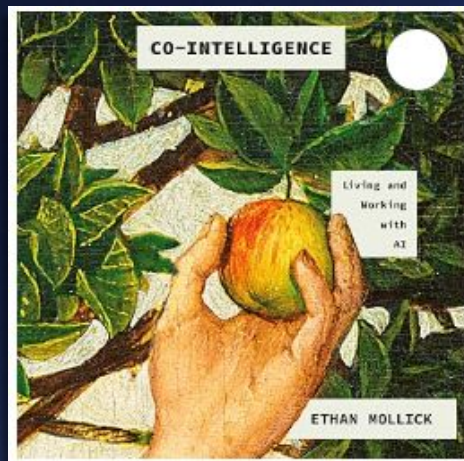


- Machine Learning A-Z
- Deep Learning A-Z
- Artificial Neural Network for Regression
- AI Engineering Core Track: LLM Engineering, RAG, QLoRA, Agents
- AI Engineering Agentic Track: The Complete Agent & MCP Course



Edgar Lobaton's Graduate Courses

- Physics Informed Machine Learning
- AI Model Deployment



Perceptions of AI

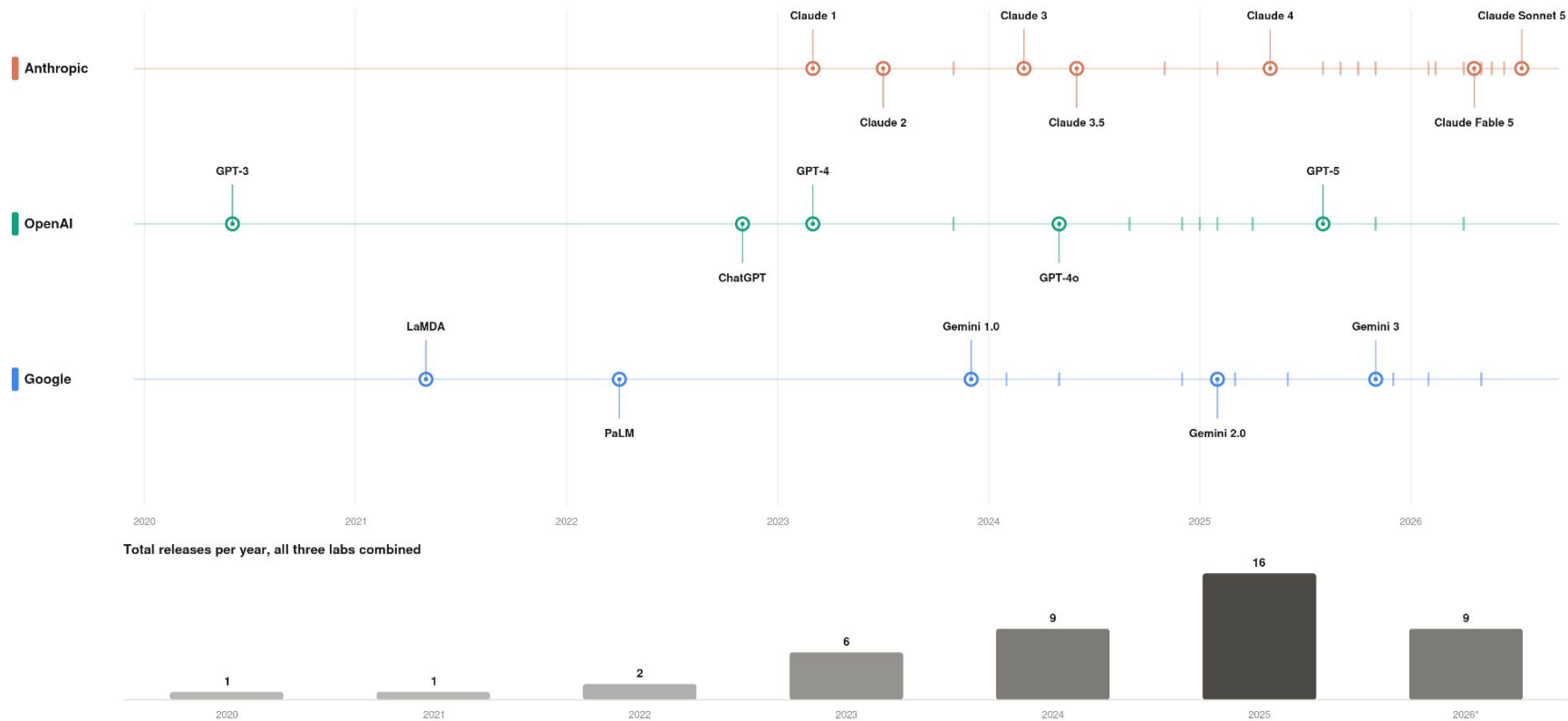
The greatest thing ever!



We're all going to die!

The Accelerating Pace of LLM Releases

Major model releases from Anthropic, OpenAI, and Google (2020–2026)



Ticks mark every public release (43 total); labeled points are generation-defining flagships. *2026 partial (through Jul 2026). Sources: OpenAI, Anthropic, Google DeepMind.

Frontier Models Are Getting Really Good

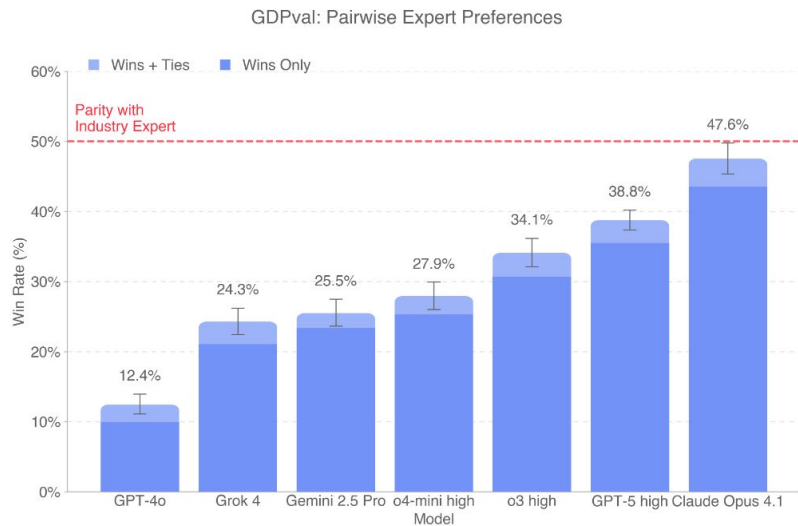
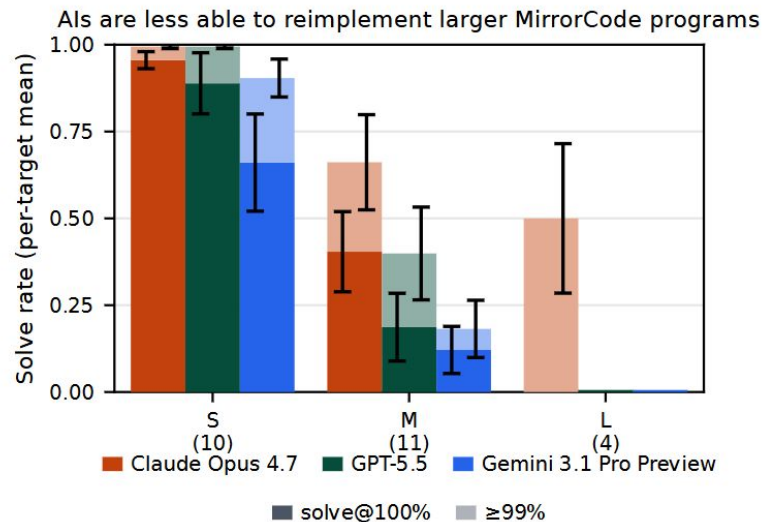


Figure 5: On human pairwise comparisons, models are beginning to approach parity with industry experts on the GDPval gold subset.

<https://cdn.openai.com/pdf/d5eb7428-c4e9-4a33-bd86-86dd4bcf12ce/GDPval.pdf>



Adamczewski, T.; Owen, D.; Rein, D.; Brand, F.; Edkins, G.; Hart, A.; O'Connell, D. MirrorCode: AI Can Rebuild Entire Programs from Behavior Alone. arXiv June 29, 2026. <https://doi.org/10.48550/arXiv.2606.30182>.

Play

My Timeline with LLMs

LLM Prompting

ChatGPT
Gemini
Local Models (w/
Ollama)
[Prompt Engineering]

Tools with LLMs

Perplexity.AI
Elicit
NotebookLM
Local AI Models

Coding with LLMs

Google Colab
Claude Code
Gemini CLI
Codex

Agentic AI

Claude CoWork
OpenClaw
Hermes
OpenCode

Where it started - prompting

Where should we start?

+ Write a python script to...

Thinking ▾



Working with Gemini in Google Colab

The screenshot shows a Google Colab notebook interface. The title bar indicates the notebook is named "Processing Ogawa Data.ipynb". The menu bar includes "File", "Edit", "View", "Insert", "Runtime", "Tools", and "Help". The toolbar shows "Commands", "Code", "Text", "Run all", and "Connect".

The notebook content includes a section titled "Town of Cary Photocatalytic Pavement Project" with a sub-section "Ogawa Data Processing". A text cell states: "I'm using a trimmed down excel sheet."

The code cell contains the following Python code:

```
[1] import pandas as pd
from google.colab import files
## Town of Cary Photocatalytic Pavement Project
## Ogawa Data Processing

# Step 1: Upload Excel file
uploaded = files.upload()

# Step 2: Get the filename
filename = list(uploaded.keys())[0]

# Step 3: Read the specific worksheet "Ogawa Data"
df = pd.read_excel(filename, sheet_name='Ogawa Data')

# Optional: Clean up the column names for easier access in code
df.columns = df.columns.str.strip().str.replace(r'\\s+', '_', regex=True).str.replace('[^0-9a-zA-Z_]', '')

# Step 5: Display first few rows
df.head()
```

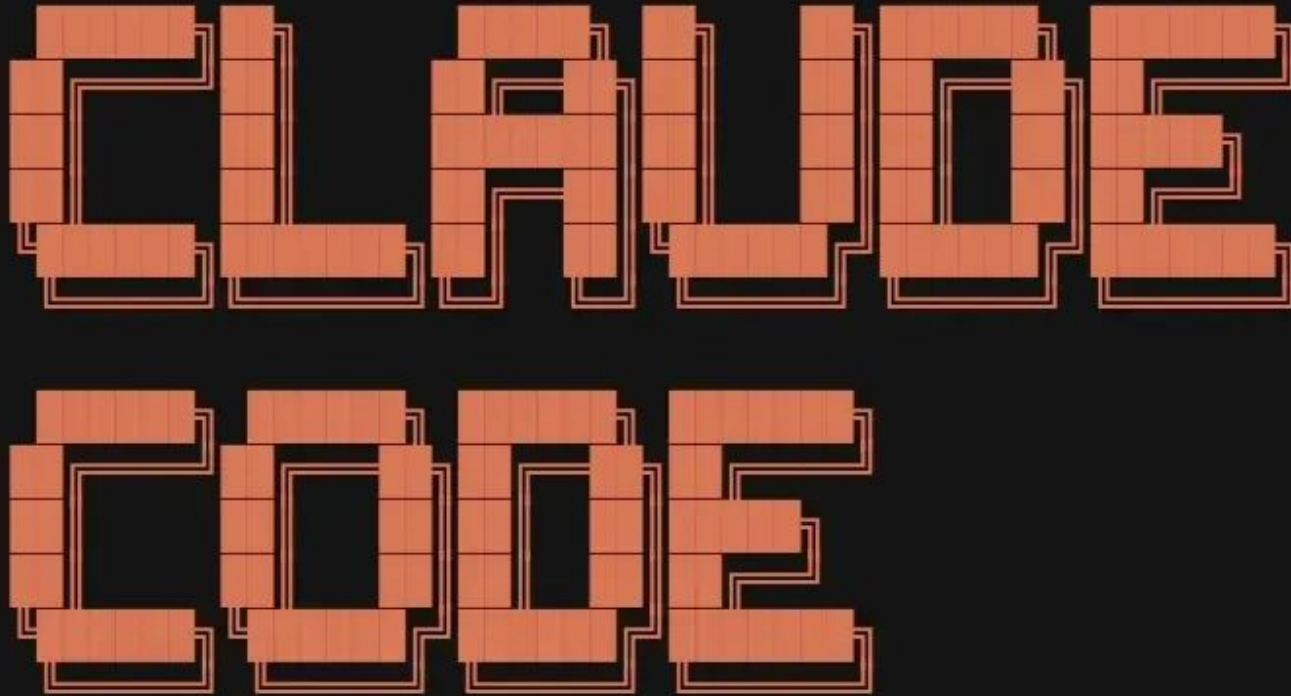
Below the code cell, a message states: "Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable." Below this message is a table of data:

	Sample_Location	Date_Deployed	Time_Deployed	Date_Recovered	Time_Recovered	Sample_Code	Sample_Type	Avg_Temperature_(°C)	Avg_Relative_Humidity_(%)	NO2_(ppb)	NOx_(ppb)
0	RGFM	2023-03-27	16:19:00	2023-04-04	09:24:00	RGFM-L-1-040423	SAMP	15.4	65.2	2.121384	4.012759
1	RGFM	2023-03-27	16:19:00	2023-04-04	09:24:00	RGFM-L-2-040423	SAMP	15.4	65.2	2.145154	4.221541
2	RGFM	2023-03-27	16:07:00	2023-04-04	09:22:00	RGFM-R-1-040423	SAMP	15.1	67.3	2.396037	3.900144
3	RGFM	2023-03-27	16:11:00	2023-04-04	09:22:00	RGFM-R-2-040423	SAMP	15.1	67.3	2.607053	4.127583
4	RGFM	2023-03-27	17:20:00	2023-04-04	02:09:00	RGFM-B-1-040423	FDBL	21.1	43.8	NaN	NaN

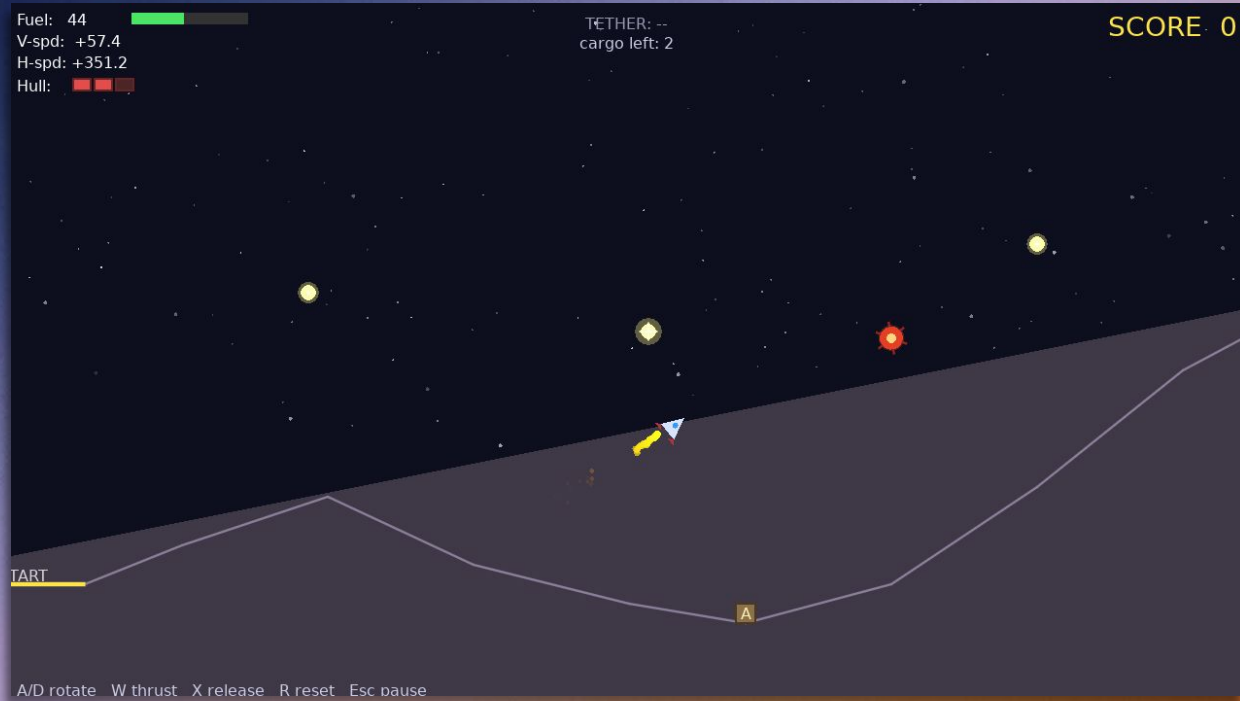
Local LLMs as callable *simple* brains



The Next Step - Coding Agents



A Simple 2D Arcade Game - Made with Claude Code



Local Models - Recursive LLM Interface

Recursive LLM Interface

Theme Examples Help

Generation settings

Default model: gemma4e2b Refresh

Temp: 0.7 Seed: 42 Max tokens: Context (num_ctx): 4096 Timeout (s): 120

System prompt: You are a precise editor. You sharpen statements to be clearer, more accurate, and more honest. Respond with ONLY the revised statement – one sentence, no preamble, no explanation, no quotation marks.

Level 1

Prompt

Sharpen this statement so it is more precise and specific.

Statement: <output 3>

If the statement above is empty, use this seed instead:
"Large language models are valuable tools for engineering researchers because they speed up writing and help with coding."

Run Level 1

Output Model:

LLMs enhance engineering research efficiency by automating tasks like literature review and code generation, mandating expert human oversight to ensure technical accuracy.

Level 2

Prompt

Introduce the strongest counterpoint or missing nuance to the statement below, and fold it into a single revised statement that is more balanced and defensible.

Statement: <output 1>

Respond with only the revised statement.

Run Level 2

Output Model:

LLMs enhance engineering research efficiency through task automation, yet this efficiency is conditional upon expert human oversight because automated outputs can contain subtle inaccuracies that require critical validation.

Level 3

Prompt

Condense the statement below into one crisp sentence of at most 20 words, keeping its nuance.

Statement: <output 2>

Respond with only the revised statement.

Run Level 3

Output Model:

+ Add Level - Remove Level

Iterations: 2

Run All

Measure Tokens

Stop

Clear All

Save

Load

Processing level 3...

Local LLM Workbench (Testing Prompts and Settings)

Untitled •
Local · Ollama

PromptExamples

HelpGet CodeSaveRun Ctrl + ↵

SYSTEM PROMPT — role, tone or context

Your job is to help me write clear and professional emails. Only respond with the email back.

USER

My name is {{name}}. I want to write an email to {{Recipient}}. Here is what I want to say: {{content}}. I want the tone to be {{tone}},

+ Add messagePre-fill response

OutputRequestHistory

Copy

Subject: Update and Apology - Mariam [Your Student ID, If applicable]

Dear Dr. Aziz,

Please accept my apologies for missing class today. I wasn't feeling well and unfortunately couldn't attend.

I'm working on getting notes from classmates to ensure I don't fall behind. Thank you for your understanding.

Sincerely,

Mariam [Your Last Name]

MODELCompare

gemma3:4b

ParametersVariables (4)

{{name}}

Mariam

{{Recipient}}

Dr. Aziz

{{content}}

I was sick. Couldn't make class. I'll get notes from friends.

{{tone}}

warm

Work Play

ABET Data Viewer Artifact

Home

Course Report

Sub Outcome Lookup

Coverage Check

Assessment Schedule

Annual Assessment

PAT Report Generator

Generate ABET-ready reports from PAT exports.

PAT exports

Drop one to three CSVs (CE, CON, ENE)

Upload

200MB per file • CSV

OK CE — summaryReportCE_F20toS26_2026_07_01_08_07.csv
uploaded 2026-07-01 (2019-2026)

OK ENE — summaryReportENE_F19toS28_2026_07_01_08_07.csv
uploaded 2026-07-01 (2019-2026)

OK CON — summaryReportCON_S20toS28_2026_07_01_08_07.csv
uploaded 2026-07-01 (2019-2026)

Assessment Schedule

Workbook with CourseSubOutcomes + OutcomeDescriptions sheets

Upload

200MB per file • XLSX

OK Schedule — Assessment Schedule - 2025.xlsx
uploaded 2026-07-01

Deploy

Course Report

Course

CE 282

Year range

20192026

Programs

CE × CON × ENE ×

Markdown (.md)

Word (.docx)

PDF (.pdf)

HTML (.html)

Preview

CE 282

Fall 2019 – Spring 2024

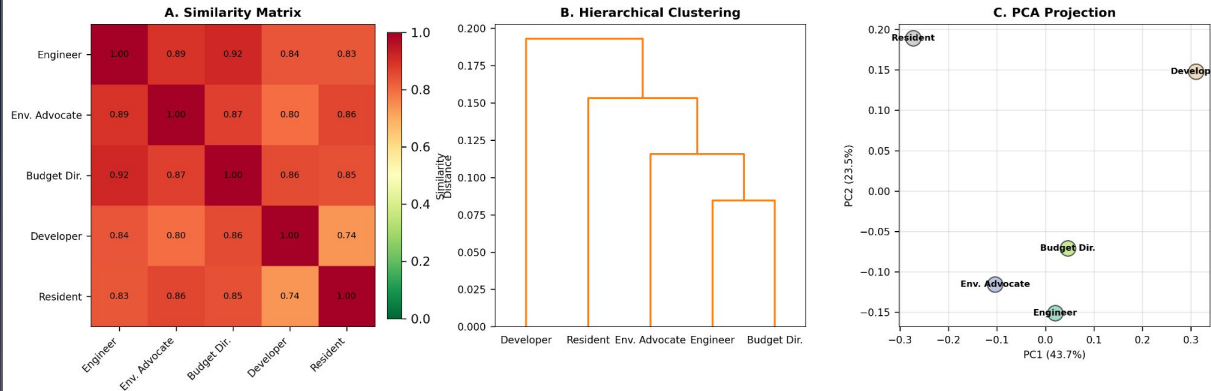
Generated July 10, 2026

Civil Engineering

Semester	Sub-Outcome	Performance Indicator	Performance
Spring 2024	1.1	50%	38%
Spring 2024	1.2	50%	61%
Fall 2021	1.1	50%	40%

AI Personas for Synthetic Stakeholder Data

Persona Differentiation Analysis: Multi-Dimensional Validation



PERSONA DIFFERENTIATION STATISTICS

Total Personas: 5

Pairwise Comparisons: 10

Distance Metrics (1 - Cosine Similarity):

Mean Distance: 0.155

Median Distance: 0.150

Min Distance: 0.085

Max Distance: 0.265

Std Deviation: 0.048

Differentiation Quality:

Well Differentiated (>0.5): 0 pairs (0.0%)

Moderate (0.3-0.5): 0 pairs (0.0%)

Poor (<0.3): 10 pairs (100.0%)

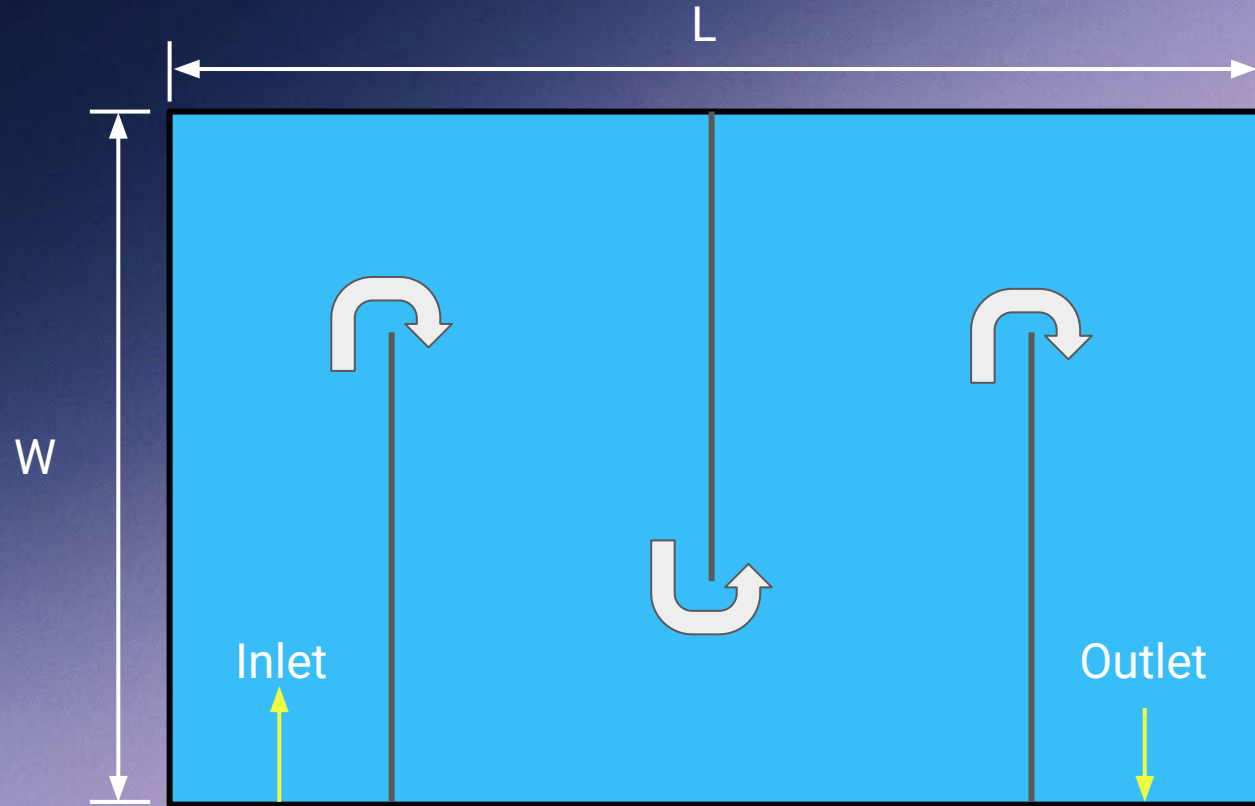
Interpretation:

✓ Average distance > 0.4 indicates good overall differentiation

✓ No pairs < 0.3 indicates all personas are distinct

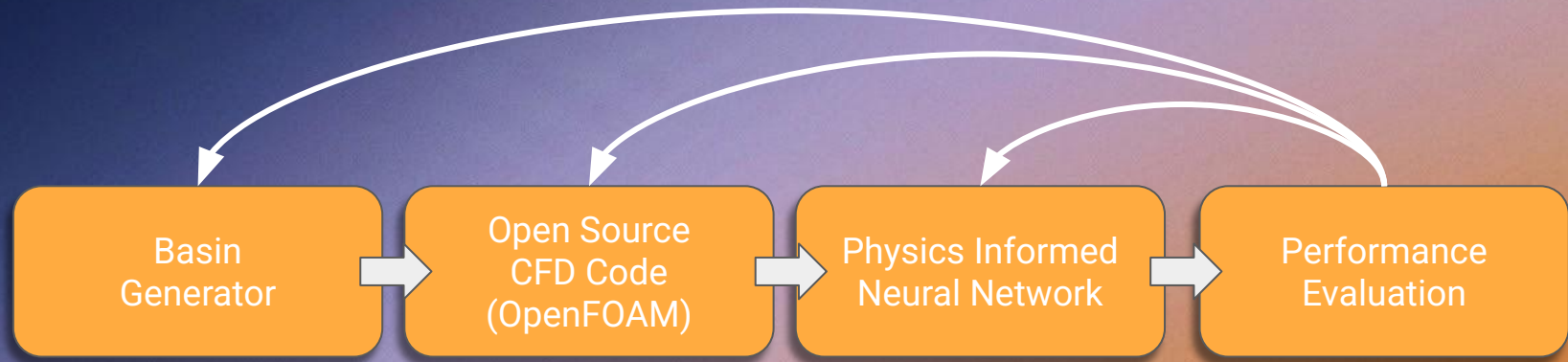
✓ Hierarchical clustering shows clear groupings by persona type

The Disinfection Contact Basin

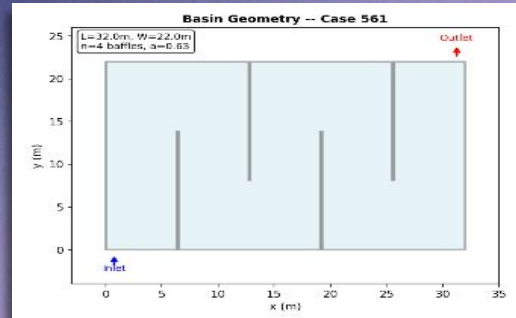


n = number of walls
 a = wall width

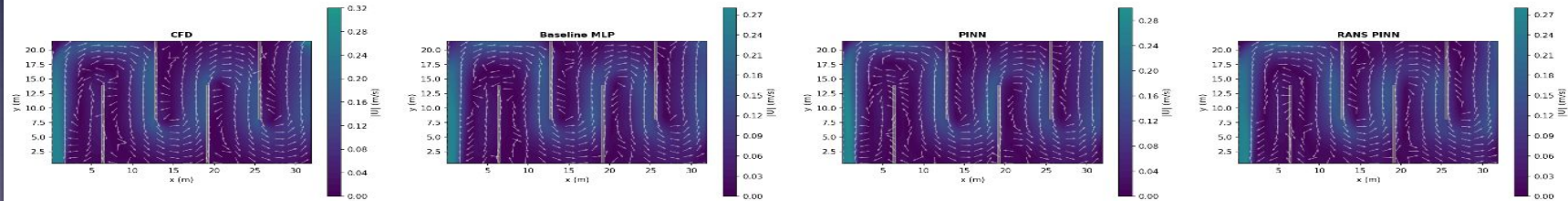
Computational Fluid Dynamics and PIML



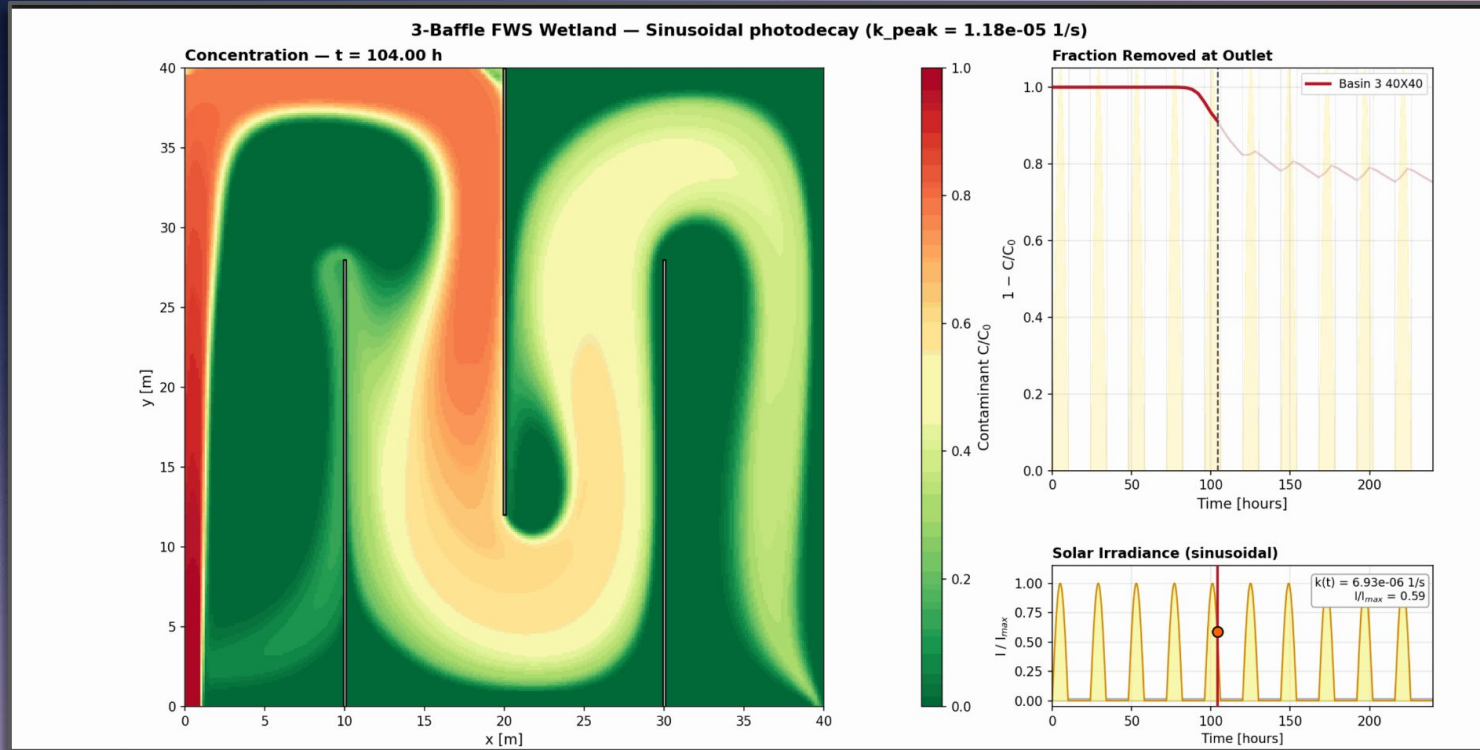
Computational Fluid Dynamics and PINNs



Velocity Contour & Continuity Comparison -- Case 561 (validation)
 $L=32.0m$, $W=22.0m$, $n=4$ baffles, $U_{inlet}=0.2607$ m/s

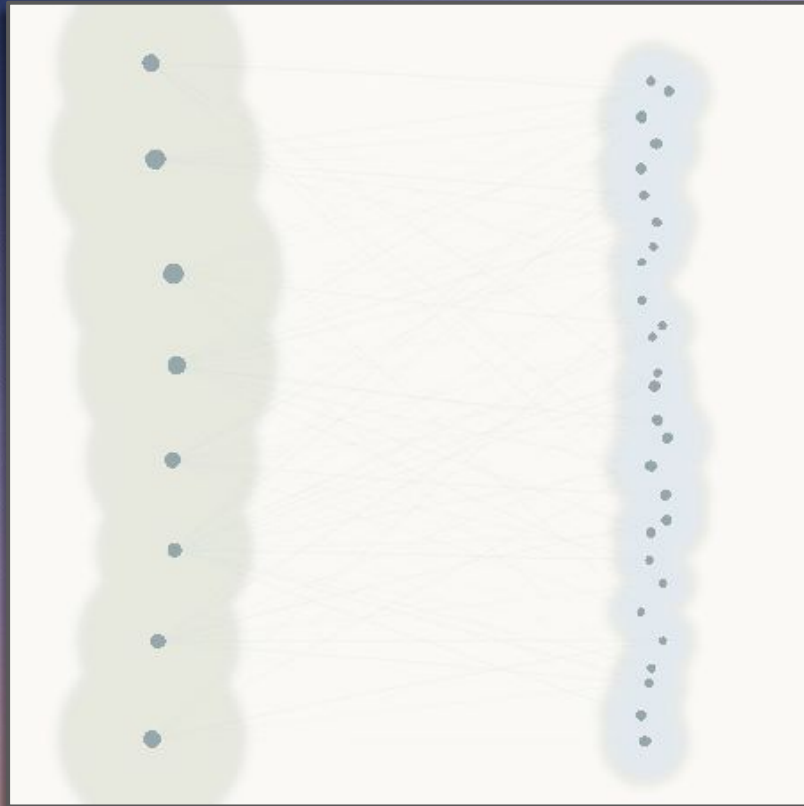


CFD + Photochemical Model for Free Surface Wetlands



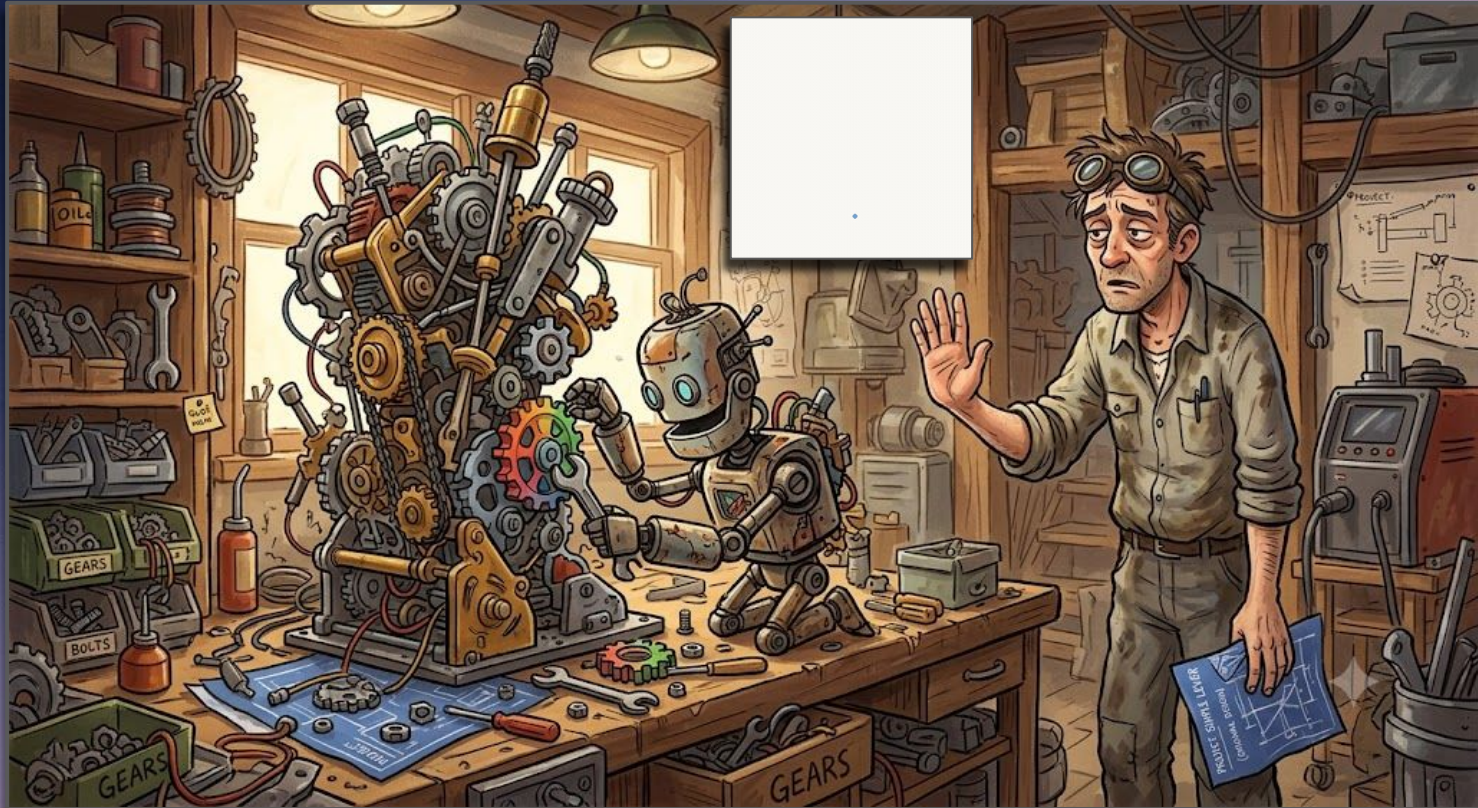
Mistakes

Assigning Judges to Students Posters as a Symposium

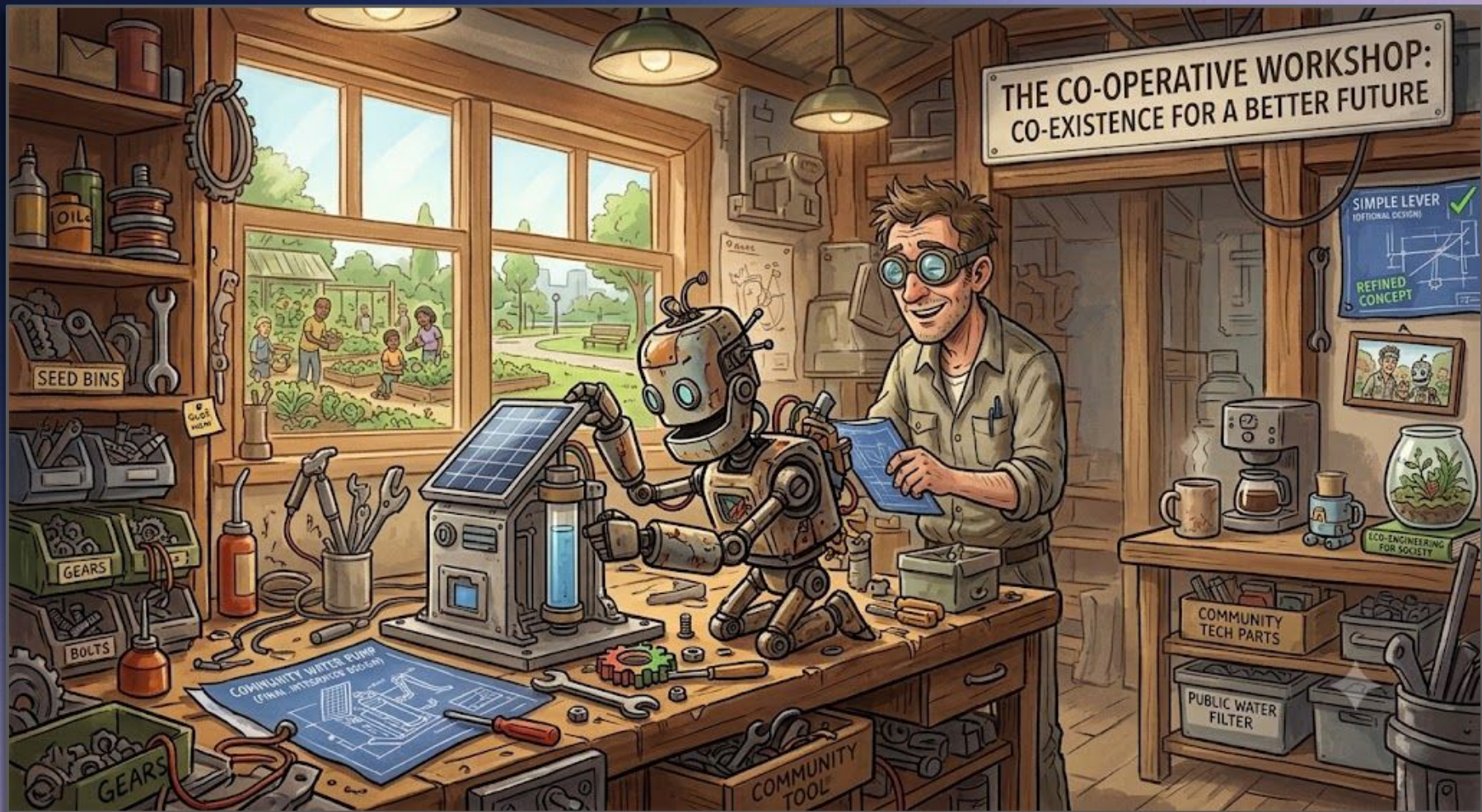


Art generated by Claude Code skill **'algorithmic art'**

Letting things get a little too complex



Co-Intelligence?



Finding Co-Intellegence

- Play with a top-tier model (e.g. Claude Opus)
- Know what you want before you ask
 - Work out a specification document before you build
- Plan with verification in mind
- Make it ask you clarifying questions
 - and ask for clarification where you don't understand
- Learn enough software craft to check its work

Thank you for your time!